

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of BSc (Microbiology) Examination April-May 2019

Course code & Name: MI 603- Basic Agricultural Microbiology

Date: 02/05/2019 Day: Thursday Time: 1:30 pm To 1:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

- Plants taken up nitrogen in form of _____.
(i) N_2 (ii) NH_4^+
(iii) NO_3^- (iv) (ii) and (iii) both
- Metarhizium anisopliae* is an insecticidal fungus that controls insects by _____.
(i) enzymatic degradation of insect cuticle (ii) secretion of antibiotic
(iii) releasing unregulated waste products (iv) all of the above
- Which of the following is an example of ecto-mycorrhiza?
(i) *Pisolithus tinctorius* (ii) *Glomus*
(iii) *Gigaspora* (iv) All of these
- Rhizosphere affects microbial population and nutrient availability by _____.
(i) exerting physical pressure (ii) secreting nutritive metabolites and organic solvents
(iii) increasing pore space (iv) changing soil structure

5. Blue green algae used as biofertilizer in _____ field.
- (i) sugarcane (ii) wheat
(iii) peanut (iv) rice
6. Slow growing rhizobium is known as _____.
(i) *Bradyrhizobium* (ii) *Sinorhizobium*
(iii) *Azorhizobium* (iv) All of these
7. Which of the following is important for the protection of nitrogenase from oxidation in *Anabaena*?
(i) expression leghemoglobin (ii) formation of heterocyst
(iii) High respiration rate (iv) None of these
8. Which media is used for the isolation of phosphate solubilizing bacteria?
(i) Pikovskayas agar (ii) Ashby's mannitol agar
(iii) Aleksandrov agar (iv) Nutrient agar
9. Which is not true about ideal carrier material used for biofertilizer and bio control agents?
(i) non-toxic to inoculant bacterial strain (ii) inexpensive
(iii) easy to sterilize (iv) non-adhesive to seeds
10. The parasitizing fungus is called hyper parasite and the parasitized fungus as hypo-parasite.
- ☐ True ☐ False

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Course code & Name: MI 603- Basic Agricultural Microbiology

Date: 02/05/2019 Day: Thursday Time: 1:50 pm To 3:30 pm Maximum Marks: 25

Instructions:

- 1. Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.
- 5.

SECTION - I

Q – II (A) Answer the following questions as directed (any two) 6

1. Write a note on “microorganisms associated with organic matter decomposition”
2. Write the difference between chemical and bio fertilizers
3. Enlist the different types of mycorrhiza and explain about its importance in agriculture.

Q – II (B) Answer the following questions. 4

1. Define: Antagonism, R:S ratio
2. Draw the labeled diagram of phosphorus cycle.

SECTION - II

Q-III Answer the following questions as directed (any three) 15

1. What is bio-control agents? Explain detail about bio-control of insects, and fungi.
2. What is diazotrophs? Explain in brief about large scale production of biofertilizer.
3. Describe the quality control measures of biofertilizers and bio control production.
4. Explain in detail about isolation and identification of rhizospheric microorganism.